Stablecoin StressBench: An 18-Event Benchmark for Execution-Aware Stablecoin Stress

Anonymous Author(s)

Abstract

Stablecoin StressBench begins from an 18-event catalogue of stablecoin stress from 2020–2023 across seven mechanisms: algorithmic reflexivity, fiat-reserve bank shock, regulatory winddown, exchange credit, DeFi pool imbalance, collateral liquidation, and RWA/niche stablecoin failure. We convert this historical universe into a tiered benchmark with source-verification status, coverage scores, machine-readable event windows, and claim-specific data permissions: Tier A supports VWAP labels and oracle bounds; Tier B supports price/liquidity claims; and Tier C supports taxonomy only. On the execution-grade SVB windows, price-visible arbitrage appears twelve times more often than it survives VWAP book-walking, fees, and settlement costs (34.3% optical versus 2.88% executable). Calm-trained models fail economically despite richer features and sequence models, while cross-mechanism meta-labeling trained on Terra/LUNA recovers 51.0% of the oracle on SVB (+82.5 bps), and a four-event mechanism-diverse protocol reaches 51.8% (+83.7 bps). A conditioned PPO-GRU agent at identical training density earns −29.2 bps, showing that reward-only policy learning does not recover the executable subset without explicit binary supervision. We release the event catalogue, source-verification registry, execution labels, oracle, model scripts, and paper tables.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

ACM Reference Format:

Anonymous Author(s). 2026. Stablecoin StressBench: An 18-Event Benchmark for Execution-Aware Stablecoin Stress. In *Proceedings of ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, 8 pages.

1 Introduction

Stablecoin stress is not a single-mechanism phenomenon. From 2020 to 2023, StressBench identifies 18 episodes spanning algorithmic death spirals, reserve-bank shocks, regulatory winddowns, exchange-credit collapses, DeFi pool imbalances, collateral-liquidation failures, and niche collateral failures. These events differ in trigger, duration, venue, affected asset, and data observability. The benchmark problem is therefore not merely to detect a depeg, but to determine which historical stress signatures remain economically executable across mechanisms.

Our thesis is that stablecoin stress is historically mechanism-diverse, but executable arbitrage is microstructure-scarce. When Silicon Valley Bank collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart. Almost none of them could be traded at a profit. VWAP slippage consumed the spread as orders cleared depth, taker fees eroded the margin, and centralised-exchange (CEX) settlement latency turned a profitable window into a loss before the transfer confirmed. Only 2.88% of one-minute windows survived all three costs and remained net-profitable. We call dislocations that appear on price charts but vanish under execution ***optical***, and the surviving fraction ***executable***. Every prior stablecoin benchmark uses the optical layer to define labels—which means every published performance number overstates what a model can actually achieve, because it is measured against windows that cannot be traded at a profit. The only empirically validated path to capturing executable dislocations is training on a mechanistically distinct stress event. We show that a model trained on an algorithmic crypto collapse (Terra/LUNA) transfers to a traditional bank run (USDC/SVB) because CEX market makers withdraw depth under uncertainty regardless of the trigger's cause—the order book does not know why liquidity is leaving, only that it is.

Stablecoin StressBench corrects this by building a historical-to-execution benchmark. The paper follows one spine: an 18-event historical stress universe, a mechanism taxonomy, a data-tier funnel, an execution-aware benchmark, cross-mechanism transfer, and an RL failure diagnostic. The catalogue defines the event universe; the tier system separates price/liquidity-grade evidence from execution-grade windows; VWAP book-walking creates per-minute executable labels; and the model ladder asks which historical stress signatures transfer to an unseen reserve-bank shock. The remainder of the paper answers four research questions: what the historical stress universe looks like, how much apparent arbitrage survives execution, whether mechanism diversity helps transfer, and why reward-only RL fails.

Contributions.

1. **Historical catalogue.** We provide a source-tracked 18-event stablecoin stress catalogue across seven mechanisms, with event-level claim provenance, verification flags, coverage scores, machine-readable windows, and source-verified paper claims.
2. **Data-tier benchmark.** We formalise Tier A/B/C claim permissions so the benchmark separates execution-grade windows, price/liquidity-grade events, and context-grade taxonomy.
3. **Execution labels.** We construct VWAP book-walking labels with route provenance, taker fees, settlement penalties, and a hindsight oracle ceiling.
4. **Empirical result.** We show a 12× optical-to-executable gap and a notional-scaling collapse: at $10K, 34.3% of SVB minutes are optically dislocated but only 2.88% are executable.
5. **Transfer and policy result.** Cross-mechanism meta-labeling recovers 51.0–51.8% of the oracle, while a conditioned PPO-GRU earns −29.2 bps despite identical positive-label density.

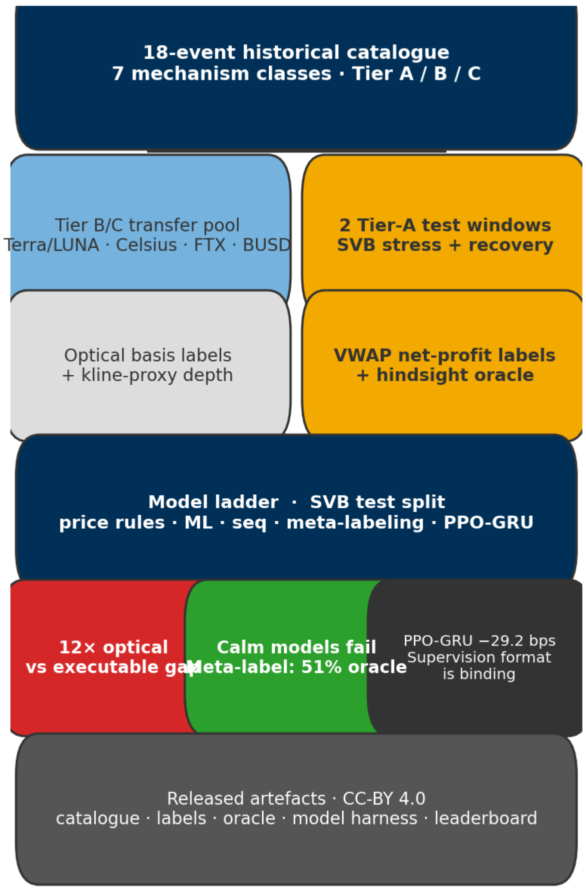

Figure 1: StressBench pipeline. The 18-event catalogue is filtered to Tier-A execution-grade windows for VWAP labels and oracle evaluation; Tier-B/C events supply mechanism-diverse transfer training. Models are ranked by mean net bps per triggered trade, not price-label AUROC.

Figure 1 shows the full benchmark pipeline. StressBench ships five artefacts: the 18-event catalogue, source-tracked event windows, execution labels with depth provenance, a hindsight oracle, and a reproducible model harness.

2 Related Work

Uhlig [1] and Griffin and Shams [2] analyse Terra/LUNA and stablecoin non-fundamental demand. We use Terra/LUNA as a cross-mechanism training split because its algorithmic trigger is mechanistically distinct from SVB's fiat-reserve failure. Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse without modelling CEX execution costs. No prior work provides per-minute executable labels or a hindsight oracle ceiling.

On limits to arbitrage: Shleifer and Vishny [4], Gromb and Vayanos [5], and Makarov and Schoar [7] show that visible price gaps need not be profitable once execution frictions are included. Our 12× optical-to-executable ratio is the same effect measured at one-minute resolution during a stress event. Hautsch et al. [3] quantify settlement latency as a first-order execution cost on blockchain assets; Cont and Kukanov [6] establish VWAP book-walking as the correct profitability measure. Gogol et al. [18] measure Maximal Arbitrage Value on AMM-to-CEX routes; our oracle extends this to CEX stress dislocations with model-evaluation benchmarking.

On ML and RL execution: López de Prado [8] introduces meta-labeling, which we extend to the cross-mechanism setting. Cintra and Holloway [9] detect the Terra depeg 12 hours early via Bayesian Online Changepoint Detection [15] on AMM reserves; we replicate their setup on CEX cross-quote data and find BOCPD achieves only AUROC 0.229, confirming that AMM and CEX signals differ fundamentally. Moallemi and Wang [16] and Ning et al. [17] supply the finite-horizon MDP framing we adopt. Mohl et al. [22] and Olby et al. [23] operate in reactive environments with market-impact feedback; our diagnostic uses historical replay without feedback, isolating training supervision rather than environment reactivity as the binding constraint.

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs, Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window, and every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 or raw futures bookDepth where archived, with proxy/partial tags where route coverage is incomplete. The buy leg (BTC-USDC) is the main challenge. The BTC-USDC perpetual contract did not exist on Binance until January 2024, so the SVB window requires kline-proxy depth, one-minute candlestick volume as a substitute for unarchived order-book depth (high-volume candles do not guarantee deep books, and a 20% haircut bounds this error). Coverage is tracked at the row and route-leg level rather than silently imputed. For Mar 10–11, incomplete BTC-USDC and USDC-USDT route legs are tagged proxy/partial; Mar 12–20 has complete cross-leg coverage in the committed route audit. All headline execution claims are recomputed under the 20% depth haircut (Table 1 gives the route-leg provenance audit). For all 2024 windows, real L2 data is available. To bound the proxy error, we apply a 20% depth haircut to the kline-proxy buy leg. The executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle remains above +130 bps. The headline result is therefore robust to the approximation.

3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 2). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	partial/proxy; real L2 where archived	real L2 (futures)
Sell BTC-USDT	raw/futures depth	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

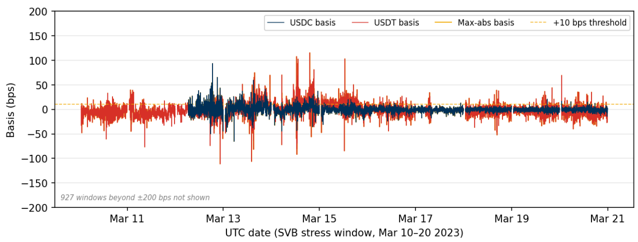

Figure 2: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.

never observed during model development. Labels are constructed by aligning each bar's outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance. We therefore report economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

3.3 Historical StressBench: Event Universe and Mechanism Coverage

Stablecoin StressBench begins with the historical event universe, not the SVB case study. The catalogue records 18 stress events from 2020–2023 and assigns each event a mechanism class, source-verification status, coverage score, event window, tier, and permitted empirical use. It is used to define the event universe, select cross-mechanism training events, assign claim types, and separate price-grade from execution-grade tasks.

Table 3 summarises the seven mechanism classes. Tier-A events support VWAP execution labels and oracle bounds; Tier-B events support price/liquidity claims only; Tier-C events contribute taxonomy coverage. The pooled transfer experiment draws from Terra/LUNA (algorithmic), Celsius/3AC and FTX (exchange credit), and BUSD (regulatory), all held mechanistically distinct from the Tier-A SVB test event (fiat-reserve bank shock).

Table 3: StressBench event matrix: 18 events, seven mechanisms.

Mechanism class	N	Example	Tier range	Role in paper
Algorithmic/reflexive	5	Terra/UST	B–C	validation, transfer source
Fiat-reserve bank shock	2	USDC/SVB	A	execution benchmark
Regulatory winddown	2	BUSD	B–C	transfer stress mechanism
Exchange credit/liquidity	3	FTX, Celsius/3AC	B	transfer and contrast
DeFi pool imbalance	3	Curve/UST, USDT/Curve	B	on-chain liquidity stress
Collateral/liquidation	1	DAI Black Thursday	B	above-peg stress
RWA/niche	2	Acala aUSD, USDR	B–C	long-tail stress coverage
Total	**18**		2A, 11B, 5C	historical stress universe

Table 4: Comparison with related benchmarks and datasets. ✓ = fully provided; ∼ = partially; — = absent.

Work	Stress event	Exec. labels	L2 prov.	Oracle	Cross-event
Cintra & Holloway [9]	✓	–	AMM	–	✓
Hernandez Cruz et al. [21]	✓	–	DEX	–	–
Gogol et al. [18]	–	∼	CEX	MAV	–
Adams et al. [19]	–	∼	CEX	–	–
StressBench (ours)	✓	✓	CEX	✓	✓

3.4 Benchmark Positioning

Table 4 positions StressBench against the closest prior work along five axes. No prior benchmark simultaneously provides execution-aware labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation.

Unlike execution simulators that optimise within known LOB environments, StressBench studies cross-mechanism transfer under sparse historical stress labels and fixed replayed liquidity.

3.5 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer.

Feature set	Description
`price_only`	5 cross-quote basis columns
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation columns
`price_book_settle`	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework from §1. The price-to-execution gap is the ratio of Layer 1 (optical) minutes to Layer 2 (executable) minutes and quantifies how misleading price-based benchmarks are. The oracle gap is the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model), quantifying how hard the prediction problem is even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

4.1 Cross-Quote Basis and Labels

The basis measures the percentage difference between BTC priced in USDC and BTC priced in real dollars. A non-zero basis signals an apparent dislocation. Three equations define the label construction.

$$b_{\text{USDC}}(t) \;=\; 10{,}000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad \text{[bps]} \tag{1}$$

$$b_{\max}(t) \;=\; \max(|b_{\text{USDC}}(t)|,\; |b_{\text{USDT}}(t)|) \tag{2}$$

$$y_{\text{basis}}(t) \;=\; \mathbf{1}[|b_{\text{USDC}}(t{+}h)| > \tau] \tag{3}$$

Equations (1)–(3) define the basis, the pooled signal, and the label. Primary task: `basis_usdc_1m_gt10bps` (τ=10 bps, h=1 min).

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q,t) \;=\; 10{,}000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \right] \quad \text{[bps]} \tag{4}$$

$$y_{\text{arb}}(q,h,t) \;=\; \mathbf{1}\left[\max_{s \in [t,\, t+h]} \text{net}(q,s) \;>\; c \right] \tag{5}$$

Here P_{ref} is the Coinbase BTC-USD mid-price and $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are per-leg taker fees and settlement latency, all as dimensionless fractions. The profit threshold is c=10 bps, taker fee 4 bps per side, and δ=5 bps. Equations (4) and (5) define the primary economic task `executable_arb_q10000_5m`. The hindsight oracle takes every minute where $\text{net}(q,t) > c$, establishing a ceiling no deployable model can exceed.

5 Models and Evaluation

5.1 Model Ladder

The ladder is organised around the benchmark thesis: price-visible stress is common, but executable opportunities are scarce and require mechanism-aware supervision. The two anchors, `NoTrade` (zero net bps) and the hindsight `Oracle`, bracket the achievable range; every other result is interpreted as a fraction of the distance between them. The price-threshold rules reproduce the implicit labels of prior optical-basis work [11]. Tabular classifiers and four nested feature sets test whether adding microstructure, fragmentation, and settlement signals closes the gap. GRU, MLP-Window, EWMA, CUSUM, and BOCPD test whether temporal or changepoint structure substitutes for stress training. Meta-labeling [8] and PPO-GRU [16, 17] then isolate supervision format: same Terra/LUNA density, explicit binary labels versus sparse reward.

Under calm training, richer feature sets do not rescue P&L; all four sets produce negative net bps because the training split contains zero stress-regime examples. Under cross-mechanism training, depth and spread features become predictive because the model has seen depth withdrawal.

5.2 RL Execution Policy Baseline

This subsection asks whether sequential decision-making can learn a profitable policy given the correct training density, by framing the entry-timing problem as a finite-horizon Markov Decision Process (MDP). At each minute t the agent observes state s_t, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q,t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.3 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.2.

5.3 Calibration and Evaluation

Net bps denotes realised mean net basis points per triggered trade after VWAP book-walking, taker fees, and settlement penalties; zero trades are reported as NoTrade rather than an undefined average. Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i > \theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. For cross-mechanism meta-labeling, Terra/LUNA is treated as a blocked training-and-calibration event: the secondary classifier is fit on the training block, the decision threshold is selected on the calibration block, and SVB remains untouched until final evaluation. The primary metrics are mean net bps per triggered trade and oracle capture percentage, defined as model mean net bps divided by oracle mean net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC (area under the ROC curve) and AUPRC (area under the precision-recall curve). `NoTrade` at 0 bps is the economic floor.

All results are reported on the SVB test split (N=15,832) with the cost parameters stated above. The kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1. Five explicit checks guard against leakage and evaluation artefacts:

(1) Random label shuffles produce zero or negative P&L for all models.
(2) Shifting features forward by one minute to simulate lookahead does not improve performance.
(3) The calm-control training split contains exactly zero executable arbitrage windows.
(4) Train and test events are separated by event identity; no SVB features appear during model development.
(5) Cross-mechanism threshold calibration uses only the held-out Terra/LUNA calibration block; the SVB test split is held out until final evaluation.

Table 5 summarises the five evaluation tasks shipped with StressBench v1.0.

Table 5: StressBench v1.0 benchmark card.

Task	Train	Val.	Test	Primary metric
`optical_basis`	calm	Terra	SVB	AUROC, AUPRC
`exec_arb_q10k`	calm	Terra	SVB	net bps, oracle cap.
`exec_arb_q50k`	calm	Terra	SVB	net bps
`cross_mech`	Terra block	Terra calib.	SVB	net bps, oracle cap.
`policy_entry` (RL)	Terra block	Terra calib.	SVB	net bps, trades

A valid StressBench submission must report mean net bps per triggered trade, trades, hit rate, oracle capture, AUROC/AUPRC, cost assumptions, notional size, calibration split, and whether the model uses Tier-A, Tier-B, or source-tracked events. Models are ranked by net bps and oracle capture, not by AUROC alone.

6 Results

6.1 The 12× Execution Barrier

The first gap is between what a price chart shows and what an order book will fill. At a 10 bps threshold, 34.3% of test minutes are optically dislocated but only 2.88% are executable after VWAP slippage, taker fees, and settlement latency. Twelve times more arbitrage appears than can be traded, and the ratio widens further with institutional trade size (Table 7). This ratio is not a threshold artefact and persists across all tested basis levels and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 4 shows the executable rate falling from 2.88% at $10K to 0.03% at $500K, with the steepest drop between $10K and $50K. Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9×, providing directional evidence that the gap is not SVB-specific. Table 6 shows the stress/recovery split: all executable windows are in the stress phase, and the recovery phase contains zero despite 21.6% optical activity.

Table 6: Optical and executable diagnostics for SVB execution-grade windows and the Terra/LUNA validation window. SVB stress/recovery are Tier-A execution-grade; Terra/LUNA is validation/training-grade and is not used for final execution-grade claims.

Window	Min.	Opt.	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	**0.00%**	—
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

Opt. = max-abs basis > 10 bps; Exec. = net-profitable windows.
Terra/LUNA labels are used for cross-mechanism training and calibration only.

The catalogue shows that price-visible stress is not confined to reserve-bank shocks. All seven mechanism classes contain documented peg, liquidity, redemption, or collateral stress episodes (Table 3). The 12× execution gap is verified only for Tier-A events with route-complete L2 depth.

Block-bootstrap CIs (60-min blocks, 2,000 resamples) confirm robustness: executable rate 95 % CI [1.47, 4.59]%; ratio [7.6, 23.0]×; oracle [+213.9, +304.2] bps.

The gap is robust across the full cost-parameter space: sweeping taker fees from 2 to 10 bps per side and settlement penalty from 0 to 10 bps moves the executable rate only between 2.84% and 3.66%, and the ratio to the optical rate remains above 3.5× throughout. Claim boundaries and data-tier scope are stated in §5.3.

6.2 Calm-Training Failure

The second gap is between a profitable window and a model that can find it in advance. Even with execution-aware labels, every model trained on calm data loses money on the executable task.

Table 7: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

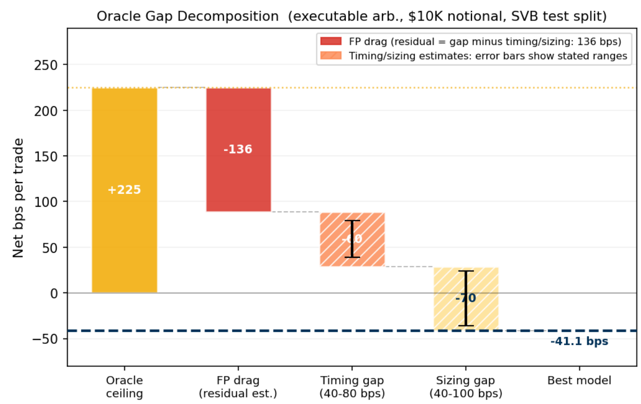

Figure 3: Oracle gap decomposition ($10K notional). Three components account for the 266 bps gap from oracle (+224.6 bps) to best model (−41 bps): FP drag, timing error, and sizing error.

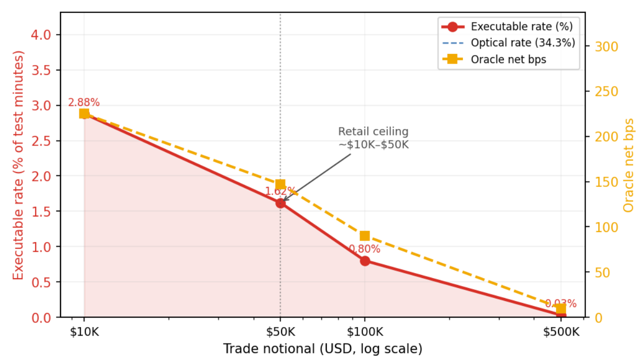

Figure 4: Notional scaling (SVB test split). The executable rate collapses from 2.88% at $10K to 0.03% at $500K: book depth, not fees, is the binding institutional constraint.

The hindsight oracle earns +224.6 bps while the best deployable model reaches −41.1 bps, a structural gap of more than 260 bps that richer features and sequence models do not close (Table 8; Figure 5).

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at −136 to −347 bps. Their mean return when wrong (−294 bps to −370 bps) reflects near-total false-positive exposure. Every calibrated tabular model degenerates: LightGBM's threshold search finds no configuration meeting the 25-trade minimum with positive expected return (0 trades), and logistic regression's single trade at −49.1 bps is not statistically meaningful. The 2.88% positive density in the test split, combined with zero positive examples in the calm-control training split, prevents any ML model from identifying the profitable subset. The `ExpNetProfitRegressor`, which directly

Table 8: Oracle gap and best-model net bps (USDC/SVB test split). *Best calibrated model is NoTrade (zero trades). ‡30-min GRU window.

Task / Model	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m`	+161.7 bps	0 bps*	162 bps	0.253
`exec_arb_q10k_5m`	+224.6 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m`	+146.8 bps	−112.7 bps	260 bps	0.063
GRU (seq)‡, `basis_usdc_1m`	+161.7 bps	−239.0 bps	401 bps	—

Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

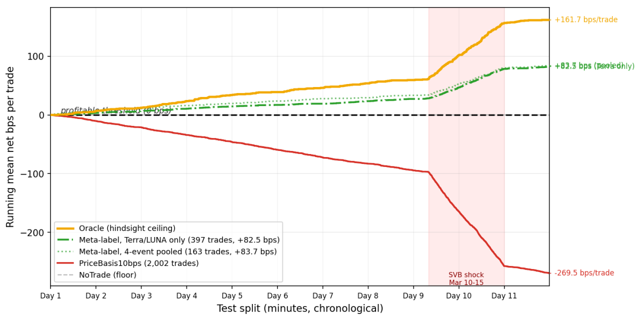

Figure 5: No calm-trained model ever crosses zero. Meta-labeling trained on a structurally different event (Terra/LUNA) is the only approach that achieves positive returns, earning +82.5 bps—roughly half the oracle's +161.7 bps. Every other learned model loses money despite sometimes high AUROC. Oracle (gold); meta-label Terra/LUNA only (green solid); meta-label 4-event pooled (green dotted); PriceBasis10bps (red); NoTrade (grey dashed). Shaded: SVB shock window.

Table 9: Tabular model ladder (`basis_usdc_1m_gt10bps`). Hit%: fraction of trades net-profitable. †Oracle ceiling. *0-trade degenerate: no threshold meets the 25-trade minimum with positive expected return.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	–	+161.7	316	–	100%
`lgbm`*	all	–	0	0.653	–
`logistic`	all	−49.1	1	0.133	0%
`gross_arb`	px	−136.0	611	0.531	4%
`price_10`	px	−269.5	2002	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	–	0.0	0	–	–
`ExpNetProfit`	px	−73.8	537	–	3%

px = `price_only`; all = `price_book_settle`.

targets the economic criterion rather than a classification proxy, reaches −73.8 bps, ruling out label mismatch as the explanation. The gap is structural (Table 9). GRU and MLP-Window sequence models confirm this, with AUROC 0.80 yet −239 bps (GRU) and −341 bps (MLP-Window). High ranking accuracy does not translate to positive returns when 97% of test windows are non-executable and the training split contains zero stress examples.

All calibrated ML models degenerate to zero trades under calm-training, equivalent to NoTrade. This section examines why the prediction gap is structural; the cross-mechanism fix that resolves it is in §6.3. Figure 3 decomposes the 260 bps gap into three measurable components. False-positive exposure is the dominant drag. `PriceBasis10bps` generates 1,969 trades, of which 1,851 are false positives. A root-cause analysis identifies route-direction mismatch as the primary failure mode: 581 of those false positives are triggered by USDT dislocations where the USDC route is simultaneously non-executable, at −38.7 bps each. True-positive windows carry a mean USDC basis of 344 bps, while false-positive windows carry just 0.56 bps and exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units), ruling out thin books as the main failure. The remaining gap splits into timing error (40–80 bps from entering 1–2 minutes late in a declining depth regime) and sizing error (40–100 bps as depth depletion shrinks the executable fraction from 2.88% at \$10K to 1.62% at \$50K). Meta-labeling's approximately 51% hit rate versus 6% for the price rule shows that cross-mechanism training substantially reduces route-direction mismatch, though a 49% false-positive rate and roughly 113 missed profitable windows remain, confining net returns to +82.5 bps.

6.3 Cross-Mechanism Transfer

The third gap is between a correct signal and a profitable policy. Cross-mechanism meta-labeling trained on Terra/LUNA is the only method that crosses zero, earning +82.5 bps and recovering half of the oracle return. A reinforcement learning agent given the same 17% training density earns −29.2 bps, which isolates explicit binary supervision as the ingredient that reward optimisation cannot replicate at low positive rates.

6.3.1 Cross-Mechanism Transfer and Training Diversity. Table 10 shows all policy results together. Meta-labeling trained on Terra/LUNA (1,559 primary-signal minutes, 17% executable density, all fitting confined to Terra/LUNA) achieves +82.5 bps on SVB (51.0% oracle capture). Four-event pooled training spans algorithmic, exchange-credit, and regulatory winddown mechanisms and reaches +83.7 bps (51.8%), confirming that mechanism-diverse supervision stabilises transfer.

Table 10: Transfer and policy results (`basis_usdc_1m_gt10bps`, `price_plus_book`, SVB test split). Mean net bps per triggered trade; threshold calibrated on Terra/LUNA validation.

Training source	Mechanism(s)	Net bps	Trades	Oracle cap.
Oracle	hindsight	+161.7	316	100.0%
Calm-control	none	0.0	0	0.0%
Terra/LUNA	algorithmic	+82.5	397	51.0%
Four-event pooled	alg. + exch. + reg.	+83.7	163	51.8%
PPO-GRU conditioned	RL on Terra	−29.2	919	−18.1%

Primary signal $|b_{\text{USDC}}| > 10$ bps; Terra/LUNA is validation/training-grade only.

Figure 7 provides direct empirical proof for why transfer works: ask-side depth collapses and spread widens identically in primary-signal windows across Terra/LUNA (algorithmic) and SVB stress (fiat-reserve), despite different triggers. CEX market makers withdraw depth under uncertainty regardless of the trigger's cause. At the order-book timescale, depth withdrawal is mechanism-agnostic, and its feature signature transfers even as its magnitude differs (5.9× for Terra/LUNA versus 12× for SVB). Figure 6 shows oracle capture across all four training conditions. FTX's credit shock

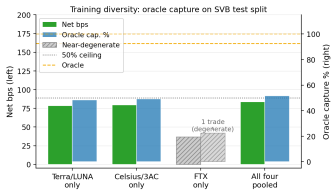

Figure 6: Oracle capture across four training conditions. Dashed: 50% ceiling; hatched: FTX near-degenerate.

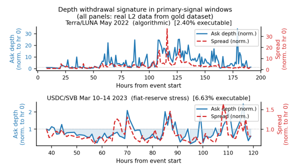

Figure 7: Depth withdrawal signature (real L2). Ask depth (blue) collapses and spread (red) widens identically across Terra/LUNA and SVB stress despite different triggers.

produced < 5 bps USDC/USDT basis at benchmark venues; its near-degenerate transfer reflects weak signal intensity, not a failure of the exchange-credit class.

Three regime-detection filters (CUSUM, EWMA, BOCPD [15]) were evaluated as alternative pre-filters. BOCPD achieves AUROC 0.229 on CEX cross-quote data: CEX basis is mean-reverting, so changepoint methods cannot substitute for the training-distribution fix that meta-labeling provides.

6.3.2 RL Policy Failure: Supervision Format, Not Label Density. This is a controlled experiment with one varying factor: how the profitable-window signal is delivered. Meta-labeling receives explicit binary labels (y=1 for every executable window); the conditioned PPO-GRU receives a sparse reward signal over the same windows. Both see identical training data at identical positive-label density (17% of primary-signal minutes are executable). Despite this, the two methods differ by 111.7 bps in net return, isolating supervision format—not density—as the binding constraint.

Without conditioning, the PPO-GRU degenerates to zero entries: the 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a calibrated reward signal. Conditioning on primary-signal windows resolves the density problem, but the agent still earns −29.2 bps on 919 trades.

The failure is qualitative, not a density problem. Meta-labeling receives explicit binary labels (y=1 for every profitable window); the RL agent must infer the same information through a reward signal negative on 83% of entries. The 919-trade overtrading pattern is consistent with a timing-transfer mismatch: the agent learned Terra/LUNA's depth-withdrawal sequence and fires on it in the SVB context even when the timing does not match. Multi-mechanism training and reactive market-impact feedback are the direct paths to closing the remaining gap.

7 Limitations and Open Questions

Each limitation below identifies a tractable next step rather than a fundamental ceiling.

Kline-proxy buy leg. The BTC-USDC buy leg uses candlestick-volume depth for the SVB and training event windows because the perpetual contract did not exist on Binance until January 2024. A 20% depth haircut confirms robustness (§6.1). The proxy dependency resolves directly with 2024 data, where real L2 is already available and partially collected.

Settlement cost assumption. The δ=5 bps penalty is validated against on-chain Etherscan data for 100,000 USDC transfers across the SVB window (Table 11). On nine of ten days the median transfer cost was 1.4–3.2 bps, well below 5 bps. The one exception is March 11 at peak depeg, where gas spiked to 61.5 gwei and the P95 cost reached 8.6 bps—the only day the assumption is tight, and even then the oracle remains profitable at +259.9 bps per trade.

Table 11: On-chain settlement cost validation (USDC ERC-20 transfers, 100K sample, Mar 10–20 2023, ETH ≈ $1,580). δ=5 bps is conservative on all days except the Mar 11 gas spike at peak depeg.

Date	Med gas (gwei)	Med cost (bps)	P95 cost (bps)	Burns
Mar 10	30.8	3.16	5.70	4
Mar 11	61.5	6.31	8.64†	16
Mar 12	21.6	2.21	3.68	22
Mar 13	26.1	2.68	4.47	29
Mar 14–19	13.9–24	1.43–2.46	2.44–4.11	0–2

†Only day where P95 cost exceeds δ=5 bps.
Gas data from 100K USDC transfers via Etherscan API.

Tier-A coverage. Execution-grade claims are currently limited to the SVB stress and recovery windows. Extending to exchange-credit or DeFi events requires route-complete L2 archives not yet available; this is the primary bottleneck for expanding the benchmark's execution-grade scope.

Release constraints and data ethics. The benchmark releases derived features, labels, event windows, source-tracking metadata, scripts, and frozen paper outputs. Some raw exchange order-book archives are API-limited, delayed, or subscription-gated; StressBench therefore records route provenance and depth-source tags so future users can distinguish reproducible derived claims from raw-data acquisition constraints. No private user-level trading data is included.

RL market-impact feedback. The conditioned PPO-GRU's −29.2 bps may reflect a timing-transfer mismatch (Terra/LUNA depth-withdrawal sequence firing at the wrong moments in SVB) rather than a fundamental RL ceiling. Pooled multi-mechanism training on the protocol established here is the direct experiment to test whether this mismatch explains the gap. Additionally, training on historical replay without market-impact feedback miscalibrates sizing in thin-book regimes, because the agent sees depth as it was rather than as it would be after its own orders clear liquidity. Integrating the cross-mechanism initialisation protocol into a reactive simulator resolves both issues simultaneously and is the natural next step toward closing the 111.7 bps gap between meta-labeling and the RL agent.

8 Conclusion

The 12× optical-to-executable gap shows that stablecoin arbitrage benchmarks built on price labels measure the wrong object. During the SVB crisis, 34.3% of minutes appeared dislocated but only 2.88% survived VWAP book-walking, fees, and settlement costs. The gap widens with trade size, making executable liquidity the binding constraint rather than price discovery alone.

StressBench also shows that executable stress is not learnable from calm data. Calm-trained rules, tabular models, and sequence models all fail economically. Meta-labeling trained on Terra/LUNA transfers to SVB and captures 51.0% of the oracle return as mean net bps per triggered trade. Four-event pooled training reaches 51.8%, confirming that mechanism-diverse supervision stabilises transfer. In contrast, PPO-GRU remains negative at −29.2 bps despite identical positive-label density, isolating explicit executable labels as the missing ingredient.

The released benchmark consists of an 18-event stress catalogue narrowed to execution-grade labels, a hindsight oracle, and a reproducible model harness, all released under CC-BY 4.0. Future work should extend Tier-A coverage, add route-aware timing labels, and integrate reactive market-impact simulation for thin-book sizing.

Acknowledgements

Anonymised for review.

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